

Datos espaciales

Sofia Chung

2025-10-17

Contents

Ejemplo de datos espaciales en R	1
Obtención de información	1
Análisis espacial	1

Ejemplo de datos espaciales en R

Sofia Chung

Obtención de información

Se obtuvo información del producto *Global Ocean Physics Reanalysis* de Copernicus (https://data.marine.copernicus.eu/product/GLOBAL_MULTIYEAR_PHY_001_030/download?dataset=cmems_mod_glo_phy_my_0.083deg_P1D-m_202311), con data en formato nc. de Temperatura Potencial Diaria a profundidad de 0.49m, y de periodo de un año correspondiente a 01/01/2021 a 31/12/2021, frente a las costas peruanas.

Análisis espacial

En primer lugar, necesitaremos los paquetes para trabajar SIG en R raster: importar, exportar y analizar datos ráster sp: analizar datos espaciales (vectoriales)

Usaremos nuestro conjunto de datos descargados de temperatura potencial a 0.49m de profundidad en nuestro repositorio, frente a las costas de Perú.

Pasamos nuestro archivo nc a RasterBrick para poder trabajar con él y verificamos la información:

```
library(ncdf4)
library(raster)
sst_sst <- raster::brick(sst)
sst_sst
```

```
## class      : RasterBrick
## dimensions  : 185, 187, 34595, 181  (nrow, ncol, ncell, nlayers)
## resolution  : 0.08333333, 0.08333333  (x, y)
## extent      : -86.87499, -71.29166, -18.54167, -3.125  (xmin, xmax, ymin, ymax)
```

```
## crs      : +proj=longlat +datum=WGS84 +no_defs
## source   : cmems_mod_glo_phy_my_0.083deg_P1D-m_1760712233161.nc
## names    : X2021.01.01, X2021.01.02, X2021.01.03, X2021.01.04, X2021.01.05, X2021.01.06, X2021.01.07, X2021.01.08, X2021.01.09, X2021.01.10, X2021.01.11, X2021.01.12, X2021.01.13, X2021.01.14, X2021.01.15, X2021.01.16, X2021.01.17, X2021.01.18, X2021.01.19, X2021.01.20, X2021.01.21, X2021.01.22, X2021.01.23, X2021.01.24, X2021.01.25, X2021.01.26, X2021.01.27, X2021.01.28, X2021.01.29, X2021.01.30, X2021.01.31, X2021.02.01, X2021.02.02, X2021.02.03, X2021.02.04, X2021.02.05, X2021.02.06, X2021.02.07, X2021.02.08, X2021.02.09, X2021.02.10, X2021.02.11, X2021.02.12, X2021.02.13, X2021.02.14, X2021.02.15, X2021.02.16, X2021.02.17, X2021.02.18, X2021.02.19, X2021.02.20, X2021.02.21, X2021.02.22, X2021.02.23, X2021.02.24, X2021.02.25, X2021.02.26, X2021.02.27, X2021.02.28, X2021.03.01, X2021.03.02, X2021.03.03, X2021.03.04, X2021.03.05, X2021.03.06, X2021.03.07, X2021.03.08, X2021.03.09, X2021.03.10, X2021.03.11, X2021.03.12, X2021.03.13, X2021.03.14, X2021.03.15, X2021.03.16, X2021.03.17, X2021.03.18, X2021.03.19, X2021.03.20, X2021.03.21, X2021.03.22, X2021.03.23, X2021.03.24, X2021.03.25, X2021.03.26, X2021.03.27, X2021.03.28, X2021.03.29, X2021.03.30, X2021.03.31, X2021.04.01, X2021.04.02, X2021.04.03, X2021.04.04, X2021.04.05, X2021.04.06, X2021.04.07, X2021.04.08, X2021.04.09, X2021.04.10, X2021.04.11, X2021.04.12, X2021.04.13, X2021.04.14, X2021.04.15, X2021.04.16, X2021.04.17, X2021.04.18, X2021.04.19, X2021.04.20, X2021.04.21, X2021.04.22, X2021.04.23, X2021.04.24, X2021.04.25, X2021.04.26, X2021.04.27, X2021.04.28, X2021.04.29, X2021.04.30, X2021.05.01, X2021.05.02, X2021.05.03, X2021.05.04, X2021.05.05, X2021.05.06, X2021.05.07, X2021.05.08, X2021.05.09, X2021.05.10, X2021.05.11, X2021.05.12, X2021.05.13, X2021.05.14, X2021.05.15, X2021.05.16, X2021.05.17, X2021.05.18, X2021.05.19, X2021.05.20, X2021.05.21, X2021.05.22, X2021.05.23, X2021.05.24, X2021.05.25, X2021.05.26, X2021.05.27, X2021.05.28, X2021.05.29, X2021.05.30, X2021.05.31, X2021.06.01, X2021.06.02, X2021.06.03, X2021.06.04, X2021.06.05, X2021.06.06, X2021.06.07, X2021.06.08, X2021.06.09, X2021.06.10, X2021.06.11, X2021.06.12, X2021.06.13, X2021.06.14, X2021.06.15, X2021.06.16, X2021.06.17, X2021.06.18, X2021.06.19, X2021.06.20, X2021.06.21, X2021.06.22, X2021.06.23, X2021.06.24, X2021.06.25, X2021.06.26, X2021.06.27, X2021.06.28, X2021.06.29, X2021.06.30
## varname   : thetao
## level     : 1
```

Vemos que la información obtenida corresponde a una clase de RasterBrick. Además, sabemos que la información anteriormente trabajada corresponde a datos diarios desde 01/01/2021 a 30/06/2021, por lo que $1 \text{ año} * 6 \text{ meses} * 30\text{-}31 \text{ días} = 181$ n layer aproximadamente ya que cada layer representa un solo día.

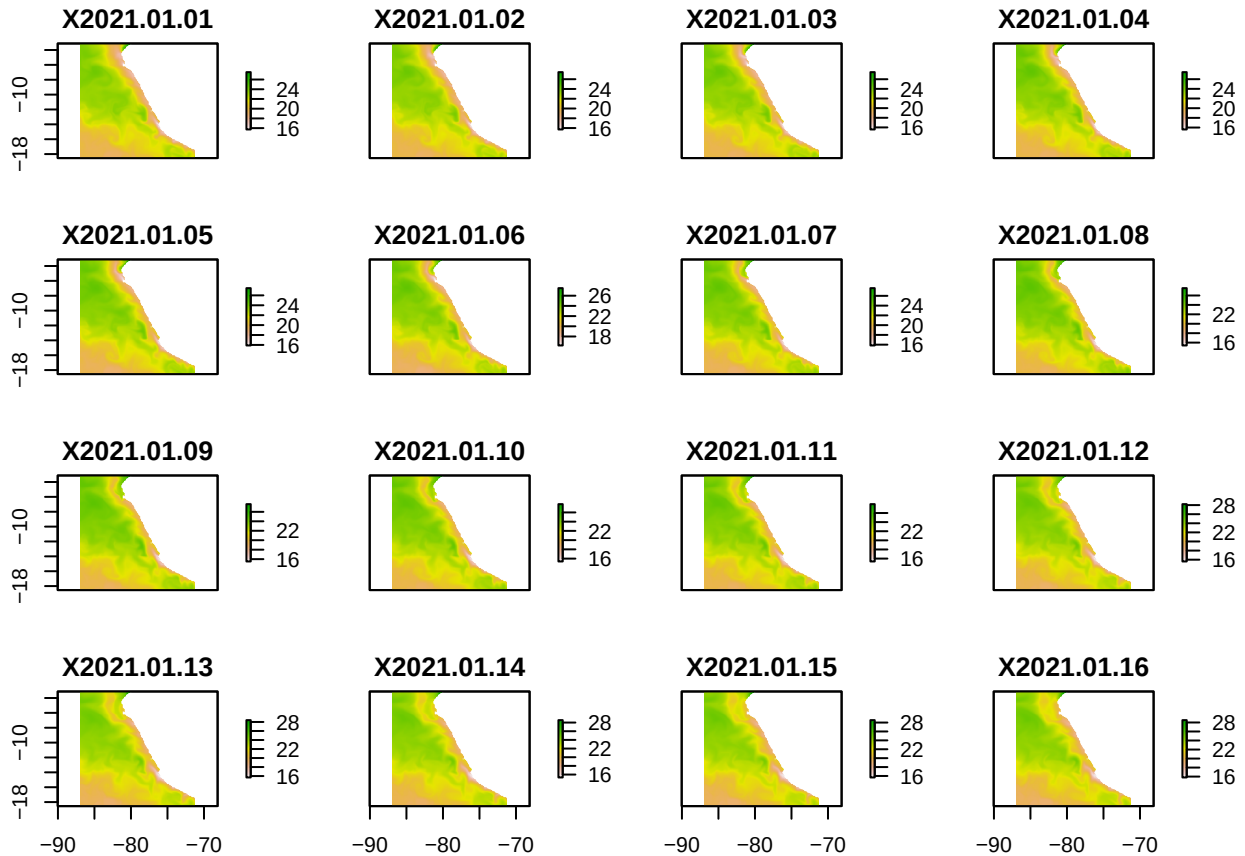
```
names(sst_sst) #para ver los layers
```

```
## [1] "X2021.01.01" "X2021.01.02" "X2021.01.03" "X2021.01.04" "X2021.01.05"
## [6] "X2021.01.06" "X2021.01.07" "X2021.01.08" "X2021.01.09" "X2021.01.10"
## [11] "X2021.01.11" "X2021.01.12" "X2021.01.13" "X2021.01.14" "X2021.01.15"
## [16] "X2021.01.16" "X2021.01.17" "X2021.01.18" "X2021.01.19" "X2021.01.20"
## [21] "X2021.01.21" "X2021.01.22" "X2021.01.23" "X2021.01.24" "X2021.01.25"
## [26] "X2021.01.26" "X2021.01.27" "X2021.01.28" "X2021.01.29" "X2021.01.30"
## [31] "X2021.01.31" "X2021.02.01" "X2021.02.02" "X2021.02.03" "X2021.02.04"
## [36] "X2021.02.05" "X2021.02.06" "X2021.02.07" "X2021.02.08" "X2021.02.09"
## [41] "X2021.02.10" "X2021.02.11" "X2021.02.12" "X2021.02.13" "X2021.02.14"
## [46] "X2021.02.15" "X2021.02.16" "X2021.02.17" "X2021.02.18" "X2021.02.19"
## [51] "X2021.02.20" "X2021.02.21" "X2021.02.22" "X2021.02.23" "X2021.02.24"
## [56] "X2021.02.25" "X2021.02.26" "X2021.02.27" "X2021.02.28" "X2021.03.01"
## [61] "X2021.03.02" "X2021.03.03" "X2021.03.04" "X2021.03.05" "X2021.03.06"
## [66] "X2021.03.07" "X2021.03.08" "X2021.03.09" "X2021.03.10" "X2021.03.11"
## [71] "X2021.03.12" "X2021.03.13" "X2021.03.14" "X2021.03.15" "X2021.03.16"
## [76] "X2021.03.17" "X2021.03.18" "X2021.03.19" "X2021.03.20" "X2021.03.21"
## [81] "X2021.03.22" "X2021.03.23" "X2021.03.24" "X2021.03.25" "X2021.03.26"
## [86] "X2021.03.27" "X2021.03.28" "X2021.03.29" "X2021.03.30" "X2021.03.31"
## [91] "X2021.04.01" "X2021.04.02" "X2021.04.03" "X2021.04.04" "X2021.04.05"
## [96] "X2021.04.06" "X2021.04.07" "X2021.04.08" "X2021.04.09" "X2021.04.10"
## [101] "X2021.04.11" "X2021.04.12" "X2021.04.13" "X2021.04.14" "X2021.04.15"
## [106] "X2021.04.16" "X2021.04.17" "X2021.04.18" "X2021.04.19" "X2021.04.20"
## [111] "X2021.04.21" "X2021.04.22" "X2021.04.23" "X2021.04.24" "X2021.04.25"
## [116] "X2021.04.26" "X2021.04.27" "X2021.04.28" "X2021.04.29" "X2021.04.30"
## [121] "X2021.05.01" "X2021.05.02" "X2021.05.03" "X2021.05.04" "X2021.05.05"
## [126] "X2021.05.06" "X2021.05.07" "X2021.05.08" "X2021.05.09" "X2021.05.10"
## [131] "X2021.05.11" "X2021.05.12" "X2021.05.13" "X2021.05.14" "X2021.05.15"
## [136] "X2021.05.16" "X2021.05.17" "X2021.05.18" "X2021.05.19" "X2021.05.20"
## [141] "X2021.05.21" "X2021.05.22" "X2021.05.23" "X2021.05.24" "X2021.05.25"
## [146] "X2021.05.26" "X2021.05.27" "X2021.05.28" "X2021.05.29" "X2021.05.30"
## [151] "X2021.05.31" "X2021.06.01" "X2021.06.02" "X2021.06.03" "X2021.06.04"
## [156] "X2021.06.05" "X2021.06.06" "X2021.06.07" "X2021.06.08" "X2021.06.09"
## [161] "X2021.06.10" "X2021.06.11" "X2021.06.12" "X2021.06.13" "X2021.06.14"
## [166] "X2021.06.15" "X2021.06.16" "X2021.06.17" "X2021.06.18" "X2021.06.19"
## [171] "X2021.06.20" "X2021.06.21" "X2021.06.22" "X2021.06.23" "X2021.06.24"
## [176] "X2021.06.25" "X2021.06.26" "X2021.06.27" "X2021.06.28" "X2021.06.29"
## [181] "X2021.06.30"
```

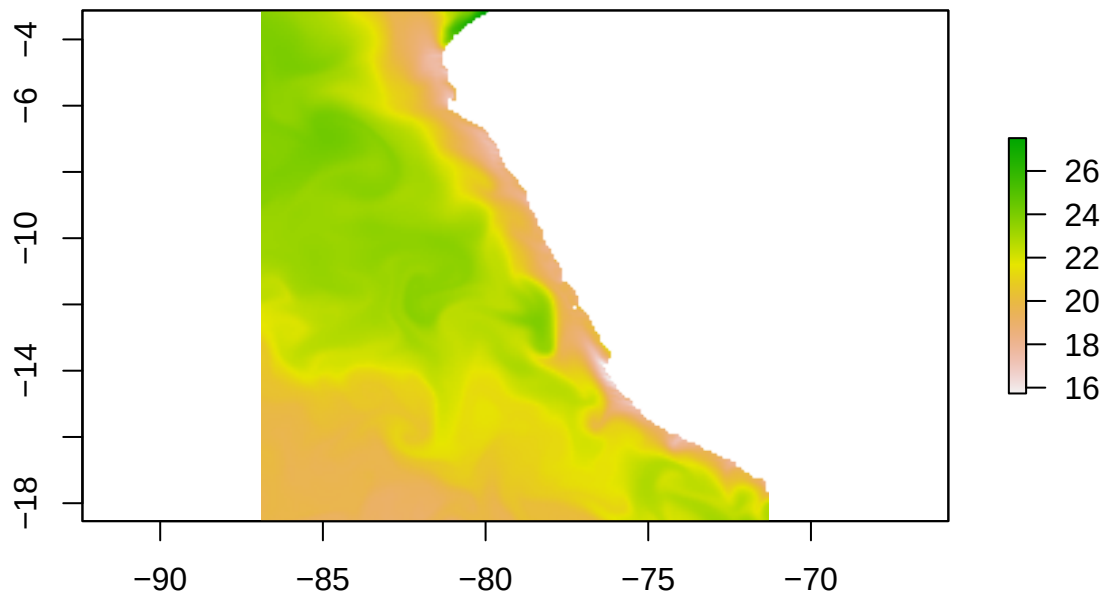
```
dim(sst_sst) #ver las dimensiones
```

```
## [1] 185 187 181
```

```
plot(sst_sst) # cada layer es un día
```



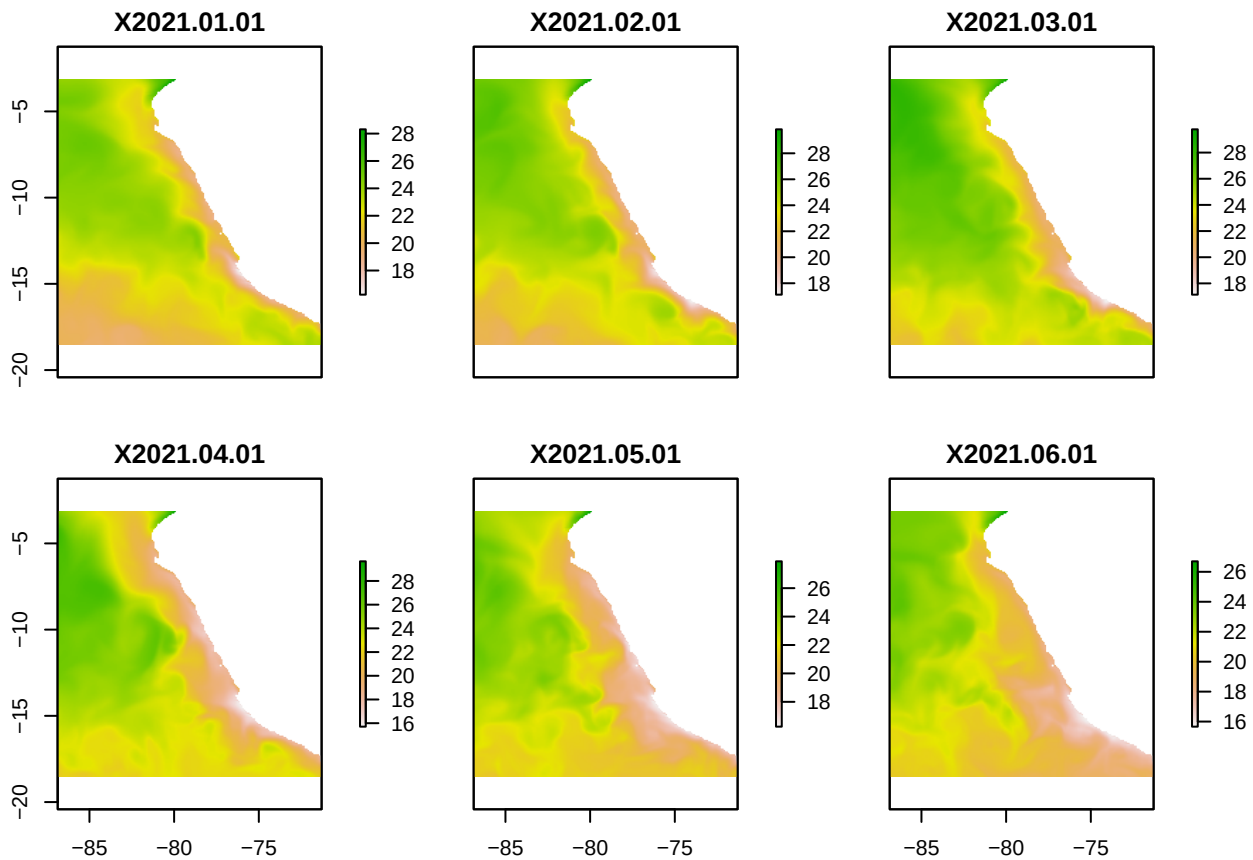
```
plot(sst_sst[[1]]) #vemos el layer del primer día (01-01-2001)
```



Ahora, con la información, podemos hacer un cálculo de promedios en función de la frecuencia temporal de nuestros datos, en este caso de diario a mensual:

```
mt_sst <- VoCC::sumSeries(sst_sst,
  p = "2021-01/2021-06", # año-mes inicial, año-mes final de mi serie de datos
  yr0 = "2021-01-01", # día inicial desde la que quiero que comience los cálculos
  l = nlayers(sst_sst), # layers de mi fichero
  fun = function(x) colMeans(x, na.rm = TRUE),
  freqin = "days", freqout = "months") # los datos entran mensuales y salen en promedio

plot(mt_sst) # un plot para cada mes
```



```
slotNames(mt_sst)
```

```
## [1] "filename" "layers" "title" "extent" "rotated" "rotation"
## [7] "ncols" "nrows" "crs" "srs" "history" "z"
```

```
nlayers(mt_sst)
```

```
## [1] 6
```

```
dim(mt_sst) #6 meses de datos
```

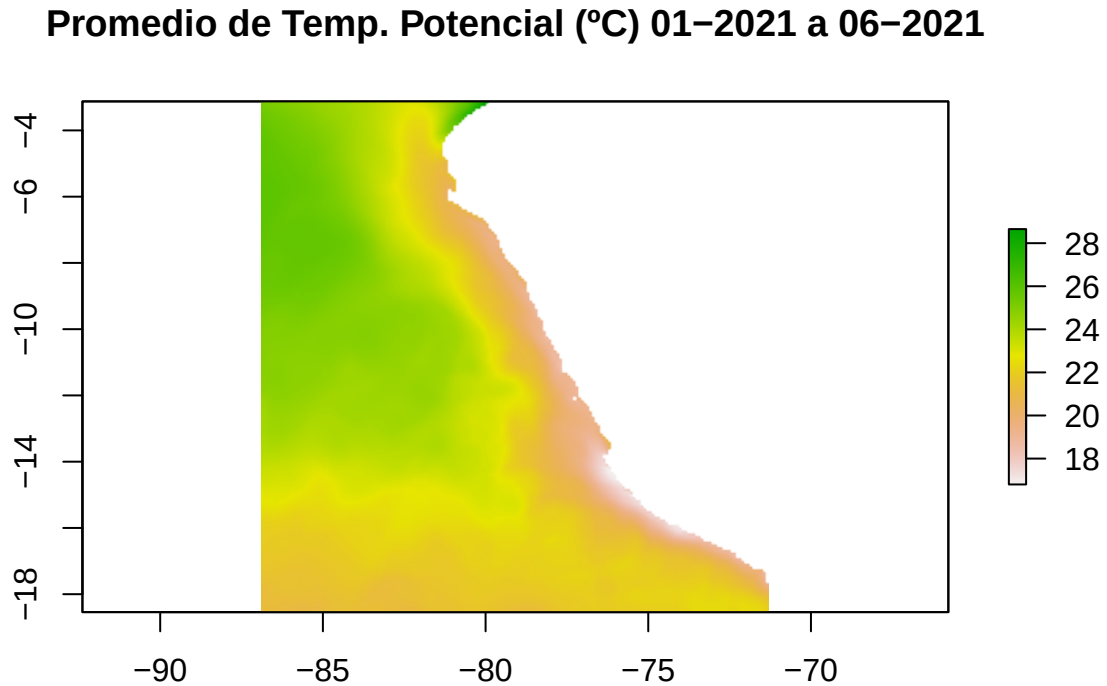
```
## [1] 185 187 6
```

```
mt_sst #resolution 1,1 1°x1°, 1°= 111 km
```

```
## class      : RasterStack
## dimensions  : 185, 187, 34595, 6 (nrow, ncol, ncell, nlayers)
## resolution  : 0.08333333, 0.08333333 (x, y)
## extent     : -86.87499, -71.29166, -18.54167, -3.125 (xmin, xmax, ymin, ymax)
## crs        : +proj=longlat +datum=WGS84 +no_defs
## names      : X2021.01.01, X2021.02.01, X2021.03.01, X2021.04.01, X2021.05.01, X2021.06.01
## min values  : 16.23831, 17.12173, 17.13770, 15.70887, 16.25355, 15.66151
## max values  : 28.31107, 29.84437, 29.77764, 29.65490, 27.88750, 26.69937
```

Con nuestros layers, ahora podemos hacer cálculo de la tendencia temporal de todos los meses en promedio:

```
#saveRDS(object = temp_trend, file = "temp_trend.Rds")
mean_all <- mean(mt_sst)
plot(mean_all, main = "Promedio de Temp. Potencial (°C) 01-2021 a 06-2021")
```



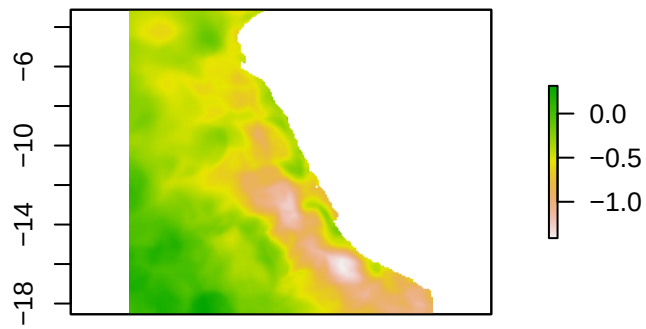
Para calcular la tendencia climática local, y obtener la pendiente, error estándar y significancia, podemos usar tempTrend

```
temp_trend <- VoCC::tempTrend(mt_sst, th = 3) # 3 nlayers slp, se y significance; 101.640 celdas
temp_trend
```

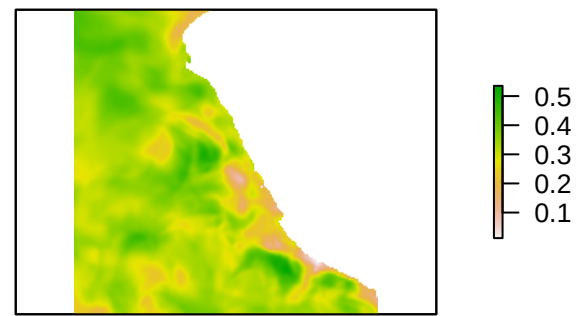
```
## class      : RasterStack
## dimensions  : 185, 187, 34595, 3  (nrow, ncol, ncell, nlayers)
## resolution  : 0.08333333, 0.08333333  (x, y)
## extent     : -86.87499, -71.29166, -18.54167, -3.125  (xmin, xmax, ymin, ymax)
## crs        : +proj=longlat +datum=WGS84 +no_defs
## names      : slpTrends,      seTrends,      sigTrends
## min values  : -1.412140e+00,  1.216364e-02,  2.099094e-06
## max values  :  0.3141399,    0.5351709,    0.9999428
```

```
plot(temp_trend, main = c("Tendencia temporal de Temp. Potencial (°C/mes)", "SE", "p-valor"))
```

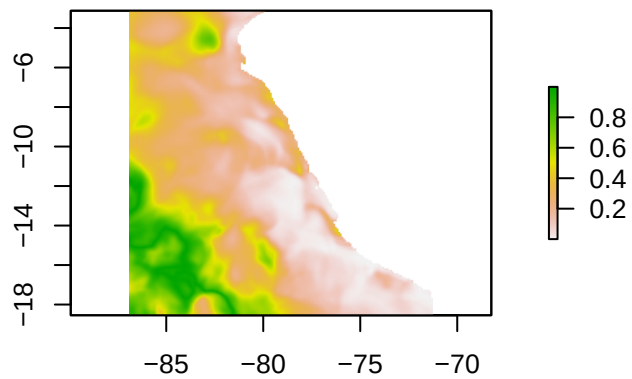
tendencia temporal de Temp. Potencial (°C/mes



SE



p-valor



```
#o podemos plotearlo de manera independiente:
# otra manera de dibujarlo
# plot(temp_trend[[1]], main = "Tendencia temporal de SST (°C/año)")
# plot(temp_trend[[2]], main = "SE Error Estándard")
# plot(temp_trend[[3]], main = "p-value")
```